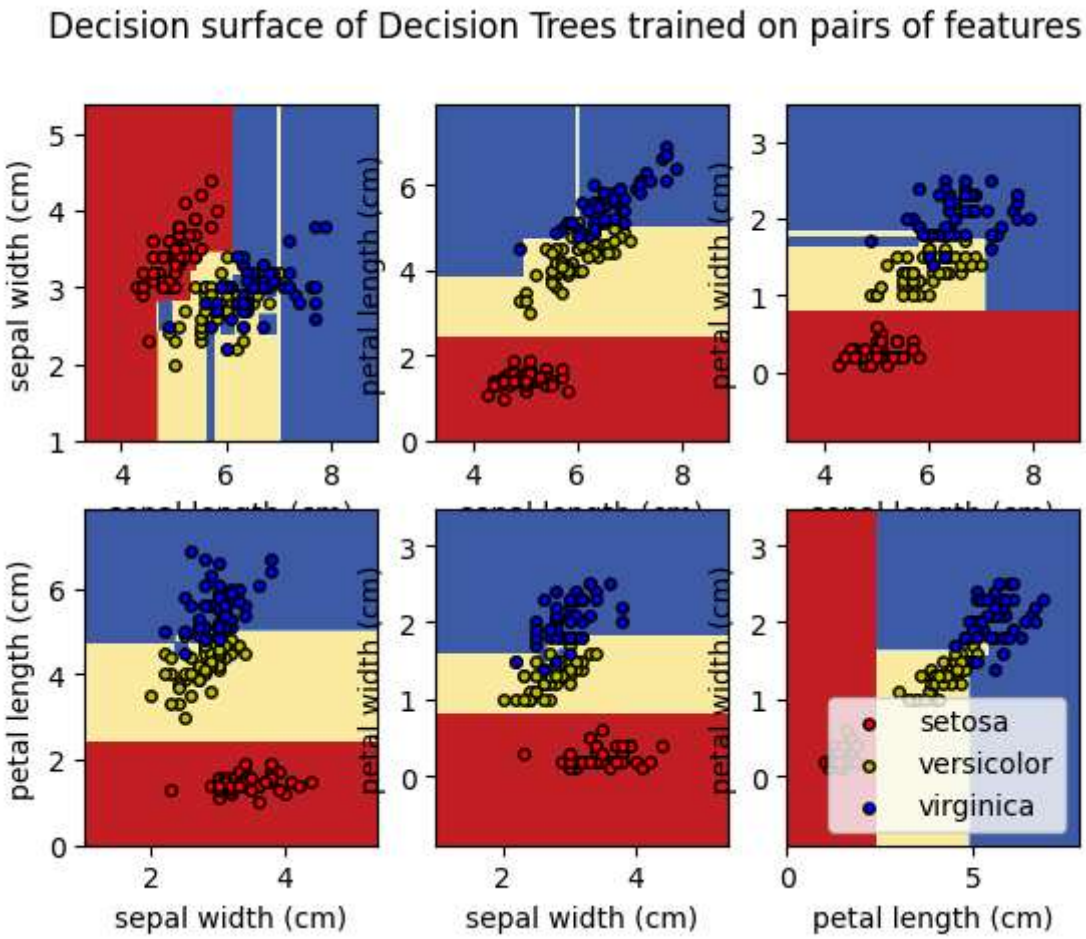
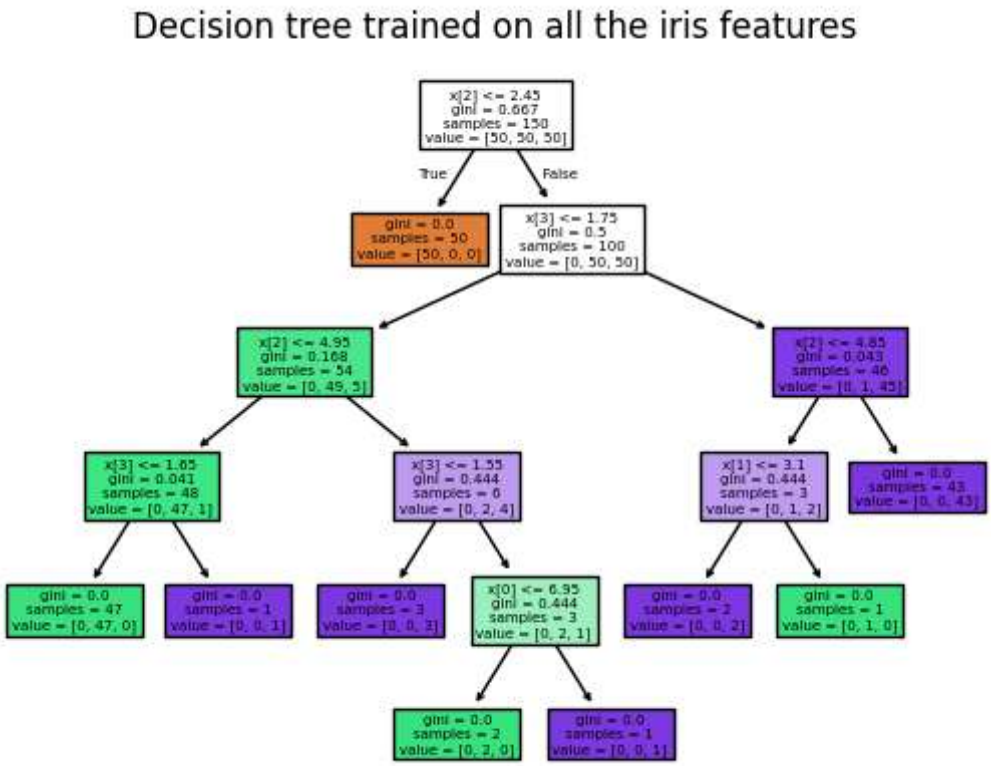
In [13]:
```
from sklearn.datasets import load_iris
import numpy as np
import matplotlib.pyplot as plt
from sklearn.tree import DecisionTreeClassifier

# Load dataset
iris = load_iris()

# Parameters
n_classes = 3
plot_colors = "ryb"
plot_step = 0.02

for pairidx, pair in enumerate([[0,1],[0,2],[0,3],[1,2],[1,3],[2,3]]):

    # We only take the two corresponding features
    X = iris.data[:, pair]
    y = iris.target

    clf = DecisionTreeClassifier().fit(X, y)

    # Plot the decision boundary
    plt.subplot(2,3,pairidx+1)

    x_min, x_max = X[:,0].min()-1, X[:,0].max()+1
    y_min, y_max = X[:,1].min()-1, X[:,1].max()+1

    xx, yy = np.meshgrid(
        np.arange(x_min, x_max, plot_step),
        np.arange(y_min, y_max, plot_step)
    )

    Z = clf.predict(np.c_[xx.ravel(), yy.ravel()])
    Z = Z.reshape(xx.shape)

    plt.contourf(xx, yy, Z, cmap=plt.cm.RdYlBu)

    plt.xlabel(iris.feature_names[pair[0]])
    plt.ylabel(iris.feature_names[pair[1]])

    # Plot training points
    for i, color in zip(range(n_classes), plot_colors):
        idx = np.where(y == i)
        plt.scatter(
            X[idx,0],
            X[idx,1],
            c=color,
            label=iris.target_names[i],
            edgecolor="black",
            s=15
        )

plt.suptitle("Decision surface of Decision Trees trained on pairs of features")
plt.legend(loc="lower right")
```

```
plt.axis("tight")
plt.show()
```



In [14]:
```
from sklearn.tree import plot_tree
plt.figure()

clf = DecisionTreeClassifier().fit(iris.data,iris.target)
plot_tree(clf, filled=True)
plt.title("Decision tree trained on all the iris features")
plt.show()
```


Decision tree trained on all the iris features
x[2] <= 2.45
gini = 0.667
samples = 150
value = [50, 50, 50]
True
False
gini = 0.0
samples = 50
value = [50, 0, 0]
x[3] <= 1.75
gini = 0.5
samples = 100
value = [0, 50, 50]
x[2] <= 4.95
gini = 0.168
samples = 54
value = [0, 49, 5]
x[2] <= 4.85
gini = 0.043
samples = 46
value = [0, 1, 45]
x[3] <= 1.65
gini = 0.041
samples = 48
value = [0, 47, 1]
x[3] <= 1.55
gini = 0.444
samples = 6
value = [0, 2, 4]
x[1] <= 3.1
gini = 0.444
samples = 3
value = [0, 1, 2]
gini = 0.0
samples = 43
value = [0, 0, 43]
gini = 0.0
samples = 47
value = [0, 47, 0]
gini = 0.0
samples = 1
value = [0, 0, 1]
gini = 0.0
samples = 3
value = [0, 0, 3]
x[0] <= 6.95
gini = 0.444
samples = 3
value = [0, 2, 1]
gini = 0.0
samples = 2
value = [0, 0, 2]
gini = 0.0
samples = 1
value = [0, 1, 0]
gini = 0.0
samples = 2
value = [0, 2, 0]
gini = 0.0
samples = 1
value = [0, 0, 1]

In [ ]: